FISH 460
Inside the Crabrooms - AB
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Modeling and Assessing Stress Response of *Hemigrapsus* sp. Inhabiting Varying Crab Pot Conditions

Abstract
Disruptive European green crabs *Carcinus* maenas (EGCs) threaten native shore crab populations *Hemigrapsus* sp. in the Pacific Northwest intertidal regions. Their capabilities to withstand a wide range of salinity and temperature fluctuations of an intertidal zone allows them to outperform native species that inhabit overlapping niches (Ens et al. 2022). Removal efforts to combat the invasive species exists in a generalist sense with high amounts of bycatch in addition to the target EGCs (Cauvel 2024). There have been documented effects on crab pots influencing the behavioral and reproductive characteristics of trapped species (Darnell et al. 2010), though none on stress response. Here we show varied physiological stress responses of native shore crabs based on their simulated trap environments. The negative impacts that traditional fukui trapping methods on the physiology of native crabs may need to be considered as these efforts continue, in order to ensure removal efforts are benefiting these species.

Introduction
European green crabs *Carcinus* maenas are one of the top 100 worst invasive species due to their prevalence as a predator, competitor, and habitat disruptor (Ens et al. 2022). They negatively impact eelgrass beds and native shore crab species. We wanted to quantify the effect that both these predators, as well as human-made predator mitigation strategies affect these shore crabs. We anticipate elevated stress levels when exposed to EGCs and wire trapping, which would in turn impact behavioral measures such as respiration levels (elevated), protein content (reduced), and righting time (elevated). The purpose of this study is to address the gap in literature on the effects that crab traps have on crab stress. Most commonly used control efforts of EGC are with Fukui wiring traps (Ens et al. 2022), which are meant to be disruptive to its environment. However, traps to mitigate bycatch impacts are being developed, such as the "crab slab" by the Tulalip tribe (Cauvel 2024). To test the efforts of this environment simulating a natural habitat to reduce the impact on bycatch species, we created replications of these pot trap types to determine how different simulated crab traps conditions affected the stress response of native shore crabs *Hemigrapsus* sp.

Methods
We obtained and labeled 12 male native shore crabs: 4 *Hemigrapsus* oregonensis (hairy shore crab) and 8 *Hemigrapsus* nudus (purple shore crab). We also obtained 1 *Carcinus* maenas

For our simulated trap conditions, we obtained 4 respirometer chambers. We developed a crab slab - inspired chamber (Cauvel 2024) by using a UV black out sheet and wrapping the container in duct tape and lining the bottom with wet sand harvested from Golden Gardens (created 2 of these). Our fukui trap - inspired chambers consisted of a mesh wiring that we cut and sewed to fit flushed against the chamber.

In 35 ppt and 24ºC water, we used two tubs that each had two respirometer chambers- one of the simulated crab slab and one of the simulated fukui trap. In one tub, we placed an EGC to roam freely. In the other, we placed a shell of similar size to the EGC to control for the visual component of the predator's presence. Each respirometer chamber consisted of three native shore crabs- two purple and one hairy to control for any variation caused by species. Into each respirometer, we connected a bubbler to keep the water oxygenated. We also changed out the water each week to prevent ammonium waste build up (Nutzhorn, Andy).

Each crab received a unique ID label super glued to its shell, was weighed, and flipped to record righting time in seconds for each of the 3 labs of data collection. Then they were placed in 6 well resazurin plates (each filled with 13 ml of resazurin assay and one crab) for 1 hour, sampling 200 microliters of the solution (and a control resazurin in a separate container) into a well plate at time intervals of 0 minutes, 30 minutes, and 60 minutes. This well plate was then run by Sarah and the data file was uploaded into the GitHub in order to measure respiration activity based on the fluorescence of the water- the more blue -> pink, the more respiration that occurred. On the final day of data collection, we performed hemolymph extractions by inserting each crab with an insulin needle. Each sample was mixed in a protein assay solution, placed in a well plate, and ran under a spectrophotometer to obtain absorbance values. Higher absorbance correlates with higher protein content of the crab.

Results

We observed no significant trends in our stress response variables- protein levels, respiration rates, and righting time. We ran Kruskal-Wallis rank sum tests because our ANOVA tests failed the normality test and could not accurately assess the variation under its assumptions.

Our protein assays resulted in a p-value 0.838 (> 0.05). There is not enough data to tell if there was an impact on protein levels due to trap type and predator presence due to no determined correlation.

The p-value determined from our respiration data was 0.781 (> 0.05). There is a slight decrease in respiration over time for groups 2 and 3. There is a large decrease in respiration over time for group 4. All three of these groups end around the same respiration, but week 1 measurements are different. Preliminary results show that crabs living in the crab slab traps or with an EGC respire more in the initial week, indicating that there might be a larger shock and stress response from being placed in these environments that decreases over time as the crabs acclimate.

Results of Kruskal-Wallis analysis for righting time determined that with and without consideration of our significant outlier of 20.37 seconds for individual 4_1 in lab 3, our p-values across treatment, time, and both were all above the alpha value of 0.05. With the outlier, p-value across treatment was 0.4704 and without it was 0.2588. Therefore there is no statistical significance in our righting time data. Our smallest p-value was produced from considering both variables of lab / time and treatment.

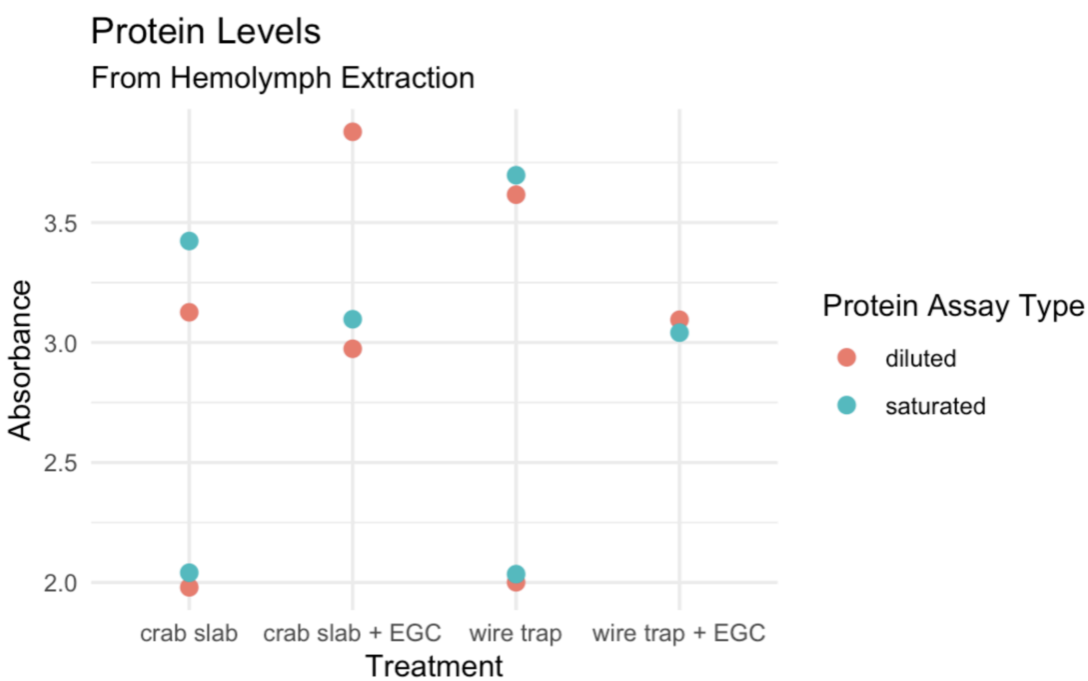

Figure 1. This figure shows protein absorbance from hemolymph extractions over the different treatments. Colors represent different assay types. Each combination of two assay types shows individual crabs. Overflows were omitted from the data and considered to be N/A.

```
        Kruskal-Wallis rank sum test

data:  num.call by treatment
Kruskal-Wallis chi-squared = 0.84615, df = 3, p-value = 0.8384
```

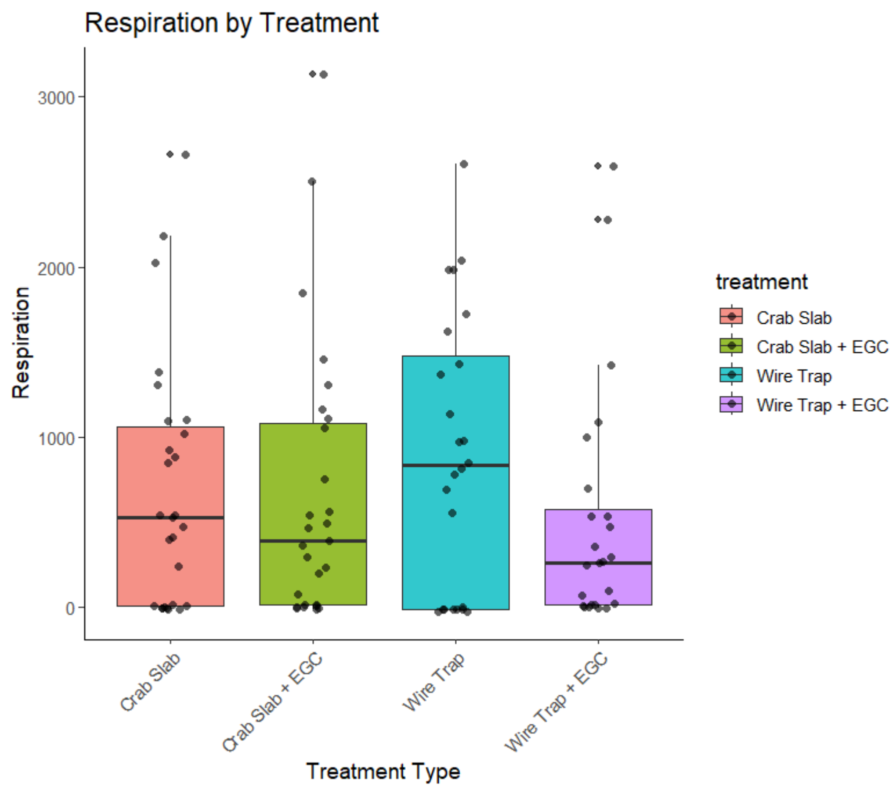

Figure 2. The figure shows the respiration level of each crab group by treatment type over the three weeks of the experiment. The colors represent each treatment and the points represent a single crabs respiration at one time point during the respiration test (3 points per crab per week).

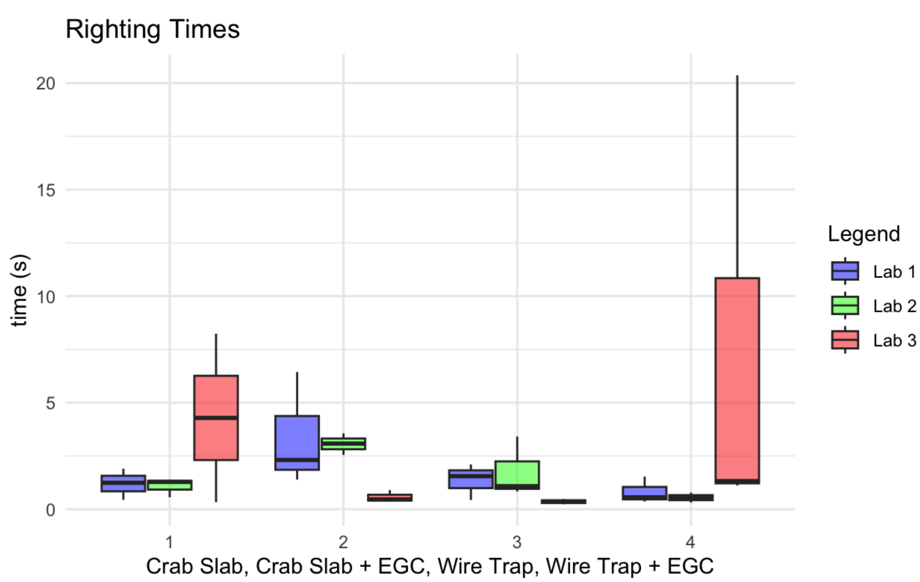

Figure 3. Righting time of each crab across 3 labs. Group 1 and group 4 experienced the greatest final increases by comparing lab 3 with lab 1. There was no notable linear trend for any of the groups, and group 2 and 3 experienced a decrease in righting time over our experiment times.

```
               Df Sum Sq Mean Sq F value Pr(>F)
lab             2  21.15  10.575   0.778  0.471
treatment       3  10.94   3.648   0.268  0.847
lab:treatment   6 108.67  18.112   1.333  0.285
Residuals      22 298.95  13.589
```

Two way ANOVA test- all p-values comparing across lab and across treatments and both are under alpha 0.05, meaning none of these variables are statistically significant. Failed assumptions.

```
── Kruskal-Wallis: Effect of Treatment ──
      Kruskal-Wallis rank sum test
data:  time_s by treatment
Kruskal-Wallis chi-squared = 2.5271, df = 3, p-value = 0.4704

── Kruskal-Wallis: Interaction (lab × treatment combined) ──
      Kruskal-Wallis rank sum test
data:  time_s by interaction(lab, treatment)
Kruskal-Wallis chi-squared = 16.788, df = 11, p-value = 0.1143

── Kruskal-Wallis WITHOUT outlier (time_s < 20) ──
      Kruskal-Wallis rank sum test
data:  time_s by treatment
Kruskal-Wallis chi-squared = 4.0244, df = 3, p-value = 0.2588


      Kruskal-Wallis rank sum test
data:  time_s by lab
Kruskal-Wallis chi-squared = 2.9534, df = 2, p-value = 0.2284
```

Table 1. Righting time summary statistics: Ranges of righting time in seconds (average ± SD.

	Lab 1	Lab 2	Lab 3
Crab Slab	1.19 ± 0.731	1.07 ± 0.447	4.29 ± 5.59
Crab Slab + EGC	3.38 ± 2.690	3.06 ± 0.505	0.573 ± 0.287
Wire Trap	1.36 ± 0.851	1.77 ± 1.423	0.355 ± 0.177
Wire Trap + EGC	0.817 ± 0.623	0.54 ± 0.235	7.60 ± 11.060

Discussion

Our findings were not statistically significant to conclude if simulated crab trap environments in the absence and presence of predatory European green crabs created additional stress on our native shore crabs. The p-value for respiration was 0.781. For protein assay it was 0.838, and for righting time it was 0.470. All three of these measurements provided data with a p-value greater than our alpha level of 0.05.

- Crab slab traps are more stressful initially → different than what we expected

- Presence of an EGC is more stressful initially → what we expected
- The combination of an EGC and being in a crab slab trap was the most stressful initially

It is interesting to note that we observed cannibalistic and aggressive behaviors in 3 instances, all in conditions in the absence of EGC predatory cues. Two crabs (individuals 10 and 70) in the wire trap condition were found partially eaten with 2 and 0 remaining legs respectively, presumably after their molting stage. Individual 50 in the crab slab condition was found on the 2nd week of data sampling torn to shreds by another crab in the shared space. The chemical cues of the EGC correlated with less of these recorded activities, possibly due to the additional stress and allocation of energy to anticipate escaping a predator rather than molting, fighting, and feeding on one another. Additionally, the two crabs that did die in the presence of EGC cues (30 in wire trap and 67 in crab slab) were notably untouched and fully intact during our removal process.

With future work under the assumption of a larger scale of resources / space provided, this project could be expanded to observe subjects in actual crab slabs and fukui traps within controlled environments. This provides a more accurate reflection of the conditions these crabs face when trapped in EGC removal effort trap systems in comparison to our simulated conditions within osmolarity chambers. Further research can contribute to the capture and removal efforts of green crabs in the Pacific Northwest by better understanding the impact this work has on native species.

Sources

Archdale, M. V., Añasco, C. P., Kawamura, Y., & Tomiki, S. (2007). Effect of two collapsible pot designs on escape rate and behavior of the invasive swimming crabs *Charybdis japonica* and *Portunus pelagicus*. *Fisheries Research*, *85*(1), 202–209. https://doi.org/10.1016/j.fishres.2007.02.008

Cauvel, K. (2024, June 19). Tribes test innovative trap in battle against invasive crab. *Northwest Treaty Tribes*. https://nwtreatytribes.org/tribes-test-innovative-trap-in-battle-against-invasive-crab/

Darnell, M. Z., Darnell, K. M., McDowell, R. E., & Rittschof, D. (2010). Postcapture Survival and Future Reproductive Potential of Ovigerous Blue Crabs Callinectes sapidus Caught in the Central North Carolina Pot Fishery. *Transactions of the American Fisheries Society*, *139*(6), 1677–1687. https://doi.org/10.1577/T10-034.1

Ens, N. J., Harvey, B., Davies, M. M., Thomson, H. M., Meyers, K. J., Yakimishyn, J., Lee, L. C., McCord, M. E., & Gerwing, T. G. (2022). The Green Wave: Reviewing the environmental impacts of the invasive European green crab (Carcinus maenas) and potential management approaches. *Environmental Reviews*, *30*(2), 306–322. https://doi.org/10.1139/er-2021-0059

Schroeder-Spain K, Fisher LL, Smee DL (2018) Uncoordinated: Effects of sublethal malathion and carbaryl exposures on juvenile and adult blue crabs (Callinectes sapidus). J Exp Mar Biol Ecol 504:1–9. https://doi.org/10.1016/j.jembe.2018.03.005

Tummon Flynn, P., Poirier, L. A., Beaulieu, G., Barrett, T. J., Cairns, D. K., & Quijón, P. A. (2024). On the rebound: Removal programs yield local-scale benefits but do not sustainably suppress populations of invasive European green crabs (Carcinus maenas). *Biological Invasions*, *26*(2), 451–469. https://doi.org/10.1007/s10530-023-03183-4